UART Transceiver Design

RTL Implementation, Verification, and Synthesis in Verilog

Project Overview

This project implements a parameterized Universal Asynchronous Receiver-Transmitter (UART) module in Verilog. Designed for robust serial communication, the architecture is divided into three distinct modules: a Baud Rate Generator, a Transmitter (TX), and a Receiver (RX). The design focuses on synthesizability, timing constraints, and handling asynchronous data streams effectively using oversampling techniques.

TARGET FPGA	LANGUAGE	SIMULATOR
Artix-7	**Verilog HDL**	**ModelSim**

Architecture Specifications

Parameter	Description	Value
Data Bits	Configurable payload size per frame	8 bits
Baud Rate	Transmission speed (parameterized)	115200 bps
Oversampling	Receiver sampling rate for data recovery	16x
Parity	Error detection protocol	None (Configurable)

Simulation & Verification

The verification environment utilizes a custom testbench to simulate real-world asynchronous communication. The testbench injects varying clock skews and framing errors to validate the robustness of the Receiver's state machine. Waveforms were analyzed using GTKWave to confirm timing accuracy.

```
     // Top-Level Module Instantiation Snippet uart_top #( .CLK_FREQ(100_000_000),
.BAUD_RATE(115200) ) uut ( .clk(sys_clk), .rst_n(sys_rst_n), .rx_serial(rx_pin),
.tx_serial(tx_pin), .rx_data_out(rx_data), .rx_ready(rx_done) );
```